Department of Mechanical and Production Engineering
Aarhus University

Take-Home Assignment Lec. 18–21

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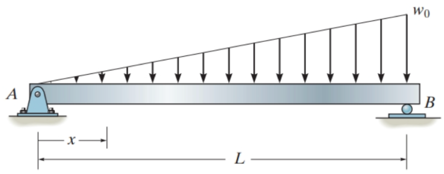
Course: Statics & Strength og Materials — 290221U006
Date: 23. November 2025

Problem 16.17

Determine the maximum deflection of the beam. EI is constant.



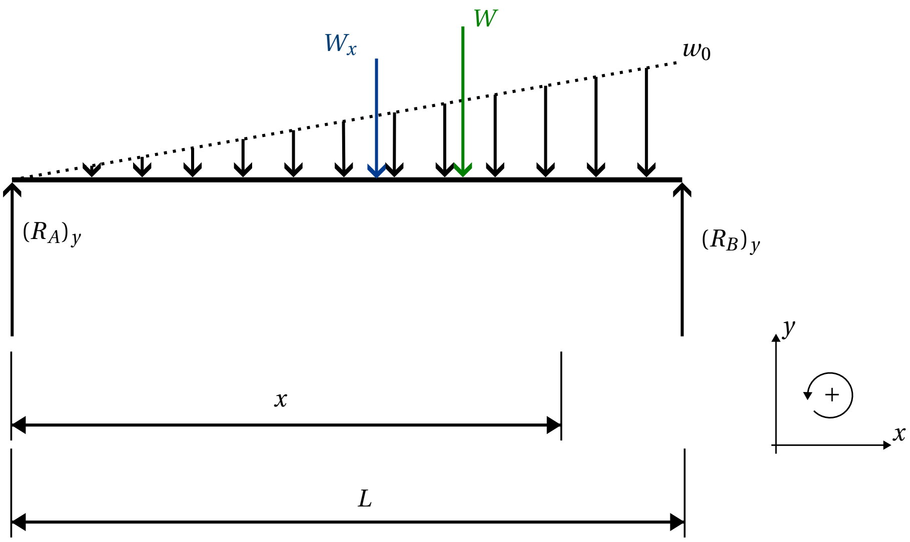


Figure 1.1: Free body diagram of the beam.

Derivation

Assuming the load varies linearly, which looks to be the case based on the figure, the weight w as a function of distance x is

$$w(x) = \frac{x}{L} w_0.$$

If we take a cut off the figure at distance x, the resultant on the left hand side ($A \to x$) is

$$W_x = \frac{1}{2}\frac{x}{L} w_0 \cdot x = \frac{x^2}{2L} w_0.$$

This will act at $1/3x$ from the cut towards A. A moment equilibrium about the cut gives

$$\begin{aligned}
0 &= \sum M_x \\
0 &= -(R_A)_y\, x + W_x \frac{x}{3} + M \\
M(x) &= (R_A)_y\, x - W_x \frac{x}{3} \\
&= (R_A)_y\, x - \frac{x^3}{6L} w_0.
\end{aligned}$$

Now we will take a moment equilibrium about point B to obtain the reaction force $(R_A)_y$. The total distributed load on the beam is

$$W = \frac{1}{2} w_0 L$$

which acts at $1/3L$ from B. Hence,

$$
\begin{aligned}
0 &= \sum M_B \\
0 &= -(R_A)_y L + W\frac{L}{3} \\
(R_A)_y &= \frac{W}{3} \\
&= \frac{L}{6} w_0.
\end{aligned}
$$

Inserting this into the expression for the internal moment as a function of distance $M(x)$ from before, we obtain

$$M(x) = \frac{w_0 L}{6} \cdot x - \frac{w_0}{6L} x^3.$$

For a beam of constant flexural rigidity (EI) a common result is

$$EI \frac{\mathrm{d}^2 v}{\mathrm{d}x^2} = M(x).$$

By insertion and subsequent integration, we obtain:

$$
\begin{aligned}
EI \frac{\mathrm{d}^2 v}{\mathrm{d}x^2} &= M(x) = \frac{w_0 L}{6} x - \frac{w_0}{6L} x^3 \\
EI \frac{\mathrm{d}v}{\mathrm{d}x} &= \int M(x)\,\mathrm{d}x = \frac{w_0 L}{12} x^2 - \frac{w_0}{24L} x^4 + C_1 \\
EI \cdot v(x) &= \int\int M(x)\,\mathrm{d}x\,\mathrm{d}x = \frac{w_0 L}{36} x^3 - \frac{w_0}{120L} x^5 + C_1 x + C_2.
\end{aligned}
$$

As the ends are fixed, our boundary conditions are

$$v(0) = 0, \quad v(L) = 0.$$

For $v(0) = 0$ to hold, C_2 must be zero. At $x = L$, we then get:

$$
\begin{aligned}
0 &= \frac{w_0 L}{36} L^3 - \frac{w_0}{120L} L^5 + C_1 L \\
\implies C_1 &= -\frac{7}{360} w_0 L^3.
\end{aligned}
$$

Hence,

$$EI \cdot v(x) = \frac{w_0 L}{36} x^3 - \frac{w_0}{120L} x^5 - \frac{7 w_0 L^3}{360} x.$$

To find the point of largest deflection we must find the extremum of this function. Hence, we set the derivative equal to zero as (note EI is ignored here, as from the above equation it can be seen to solely scale the deformation $v(x)$ and will therefore not matter when determining the point of maximum deformation):

$$
\begin{aligned}
0 &= \frac{\mathrm{d}v(x)}{\mathrm{d}x} \\
0 &= \frac{w_0 L}{12} x^2 - \frac{w_0}{24L} x^4 - \frac{7 w_0 L^3}{360}.
\end{aligned}
$$

Solving the above 4th degree polynomial gives the following 4 solutions (assuming $L \neq 0$ and $w_0 \neq 0$):

$$
\begin{aligned}
x_1 &\approx -1{,}32L \\
x_2 &\approx -0{,}519L \\
x_3 &\approx 0{,}519L \\
x_4 &\approx 1{,}32L.
\end{aligned}
$$

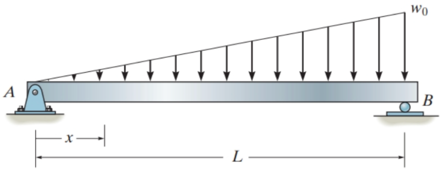
Of these, only the solution $x_3 \approx 0{,}519L$ can be the correct solution, as the other three values all lie at a point not on the beam. Substituting this solution of $x = x_3$ into our expression for $v(x)$ we obtain:

$$v_{\max} = v(x_3) \approx -6{,}52 \cdot 10^{-3} \frac{L^4 w_0}{EI}.$$

$$\boxed{v_{\max} \approx -6{,}52 \cdot 10^{-3} \frac{L^4 w_0}{EI}}$$ RESULT

Problem 16.18

Determine the maximum slope of the beam. EI is constant.



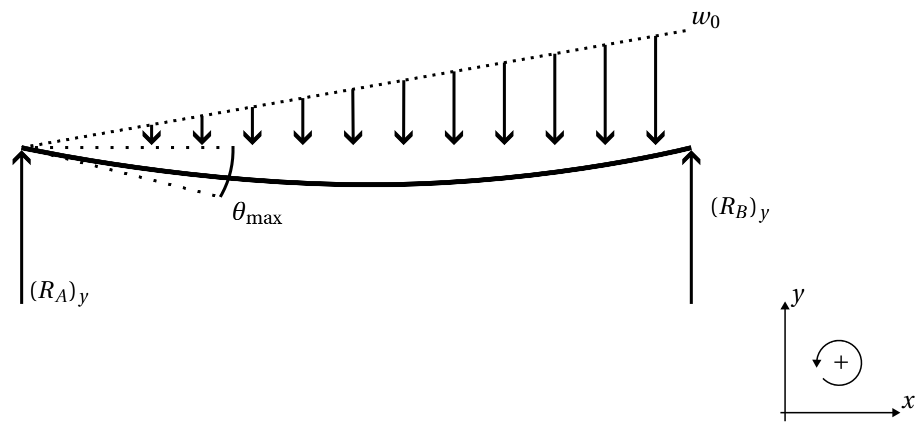


Figure 2.1: Free body diagram of the beam.

Derivation

From Problem 16.17 we have the results:

$$EI\frac{\mathrm{d}^2v(x)}{\mathrm{d}x^2} = \frac{w_0L}{6}x - \frac{w_0}{6L}x^3$$
$$EI\frac{\mathrm{d}v(x)}{\mathrm{d}x} = \frac{w_0L}{12}x^2 - \frac{w_0}{24L}x^4 - \frac{7w_0L^3}{360}.$$

As $\mathrm{d}^2v/\mathrm{d}x^2 = \mathrm{d}\theta/\mathrm{d}x$ and $\mathrm{d}v/\mathrm{d}x = \theta$ the above can also be written as:

$$EI\frac{\mathrm{d}\theta(x)}{\mathrm{d}x} = \frac{w_0L}{6}x - \frac{w_0}{6L}x^3$$
$$EI\theta(x) = \frac{w_0L}{12}x^2 - \frac{w_0}{24L}x^4 - \frac{7w_0L^3}{360}.$$

Finding the maximum slope $\theta(x)$ corresponds to finding the value of x for which $\mathrm{d}\theta(x)/\mathrm{d}x = 0$. Hence, assuming, $L \neq 0$

$$0 = \frac{w_0L}{6}x - \frac{w_0}{6L}x^3$$
$$\implies \quad x_1 = 0, \quad x_2 = L \quad .$$

Solving the equation for the slope for these values, we obtain:

$$\theta(x) = \frac{1}{EI}\left(\frac{w_0L}{12}x^2 - \frac{w_0}{24L}x^4 - \frac{7w_0L^3}{360}\right)$$

$$\theta(x_1) = \frac{1}{EI}\left(-\frac{7w_0L^3}{360}\right) = -\frac{7w_0L^3}{360EI}$$
$$\theta(x_2) = \frac{w_0L^3}{EI}\left(\frac{1}{12} - \frac{1}{24} - \frac{7}{360}\right) = \frac{w_0L^3}{45EI}.$$

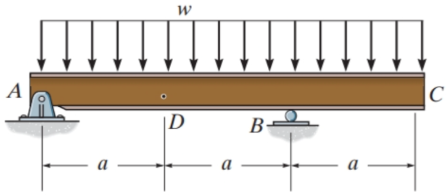
As $|-7/360| < |1/45|$ the maximum deflection occurs at $x_2 = L$, where:

$$\boxed{\theta_{\text{max}} = \frac{w_0L^3}{45EI}}$$

RESULT

Problem 16.36

Determine the slope at A and the deflection at point D of the overhang beam. EI is constant.



Derivation

As the load varies linearly with displacement (constant distributed load) and the load is not assumed large enough to significantly distort the original geometry of the beam, the method of superposition can be employed. Using this, we split up the beam into the two sections shown on XXXFBDXXX. One of these is the simply supported beam AB with the uniform load w and the other being the effect on AB from the uniform load on the overhang BC.

Simply supported beam AB

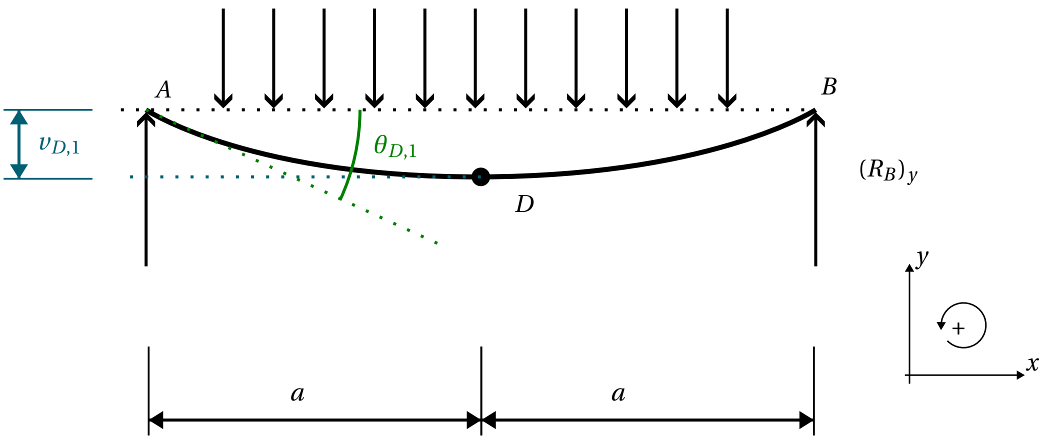


Figure 3.1: Free Body diagram of the simply supported beam AB.

For the simply supported beam, depicted on Figure 3.1, Appendix D gives the slope at A as:

$$\theta_{A,1} = -\frac{wL^3}{24EI} = -\frac{w(2a)^3}{24EI} = -\frac{wa^3}{3EI}.$$

The deflection at point D will, due to symmetry, be the maximum deflection. From Appendix D, this is seen to be

$$v_{D,1} = -\frac{5wL^4}{384EI} = -\frac{5w(2a)^4}{384EI} = -\frac{5wa^4}{24EI}.$$

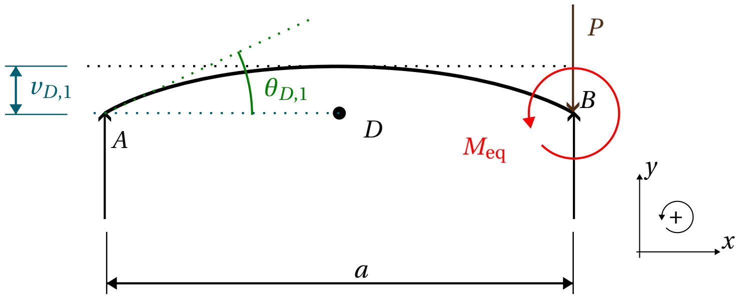


Figure 3.2: Free Body diagram of the contribution from the overhang BC.

Contribution from overhang BC

We will replace the distributed load over BC with a corresponding couple moment and a force acting downwards on R_B as shown on Figure 3.2. The equivalent moment M_{eq} is then given by

$$M_{\text{eq}} = -w \cdot a \cdot \frac{1}{2} \cdot a = -\frac{1}{2} w a^2.$$

From the Appendix D the slope at A caused by the moment from the overhang BC is

$$\theta_{A,2} = \frac{-M_0 L}{6EI} = -\frac{\left(-\frac{1}{2} w a^2\right) 2a}{6EI} = \frac{wa^3}{6EI}.$$

And the deflection is found from the elastic curve given in the Appendix as

$$v(x) = \frac{-M_0 x}{6EIL}\left(L^2 - x^2\right) = \frac{wax}{24EI}\left(4a^2 - x^2\right).$$

Evaluating this at $x = a$ gives the deflection for D as

$$v(a) = v_{D,2} = \frac{wa^2}{24EI}\left(4a^2 - a^2\right) = \frac{wa^4}{8EI}.$$

Principle of superposition

Using the principle of superposition we find the total slope at A as the algebraic sum of the slopes caused by either section, as:

$$\theta_A = \theta_{A,1} + \theta_{A,2} = -\frac{wa^3}{3EI} + \frac{wa^3}{6EI} = -\frac{wa^3}{6EI}.$$

Likewise, the total deflection at D will be the algebraic sum of the deflections at D caused by either of the two sections:

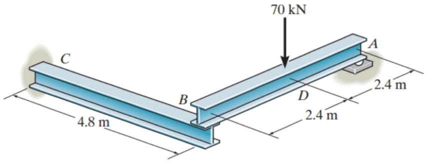
$$v_D = v_{D,1} + v_{D,2} = -\frac{5wa^4}{24EI} + \frac{wa^4}{8EI} = -\frac{wa^4}{12EI}.$$

$$\boxed{\begin{aligned} \theta_A &= -\frac{wa^3}{6EI} \\ v_D &= -\frac{wa^4}{12EI}. \end{aligned}}$$

RESULT

Problem 16.46

The assembly consists of a cantilevered beam CB and a simply supported beam AB. If each beam is made of A–36 steel and has a moment of inertia about its principal axis of $I_x = 50 \cdot 10^6\,\text{mm}^4$, determine the displacement at the center D of beam BA.



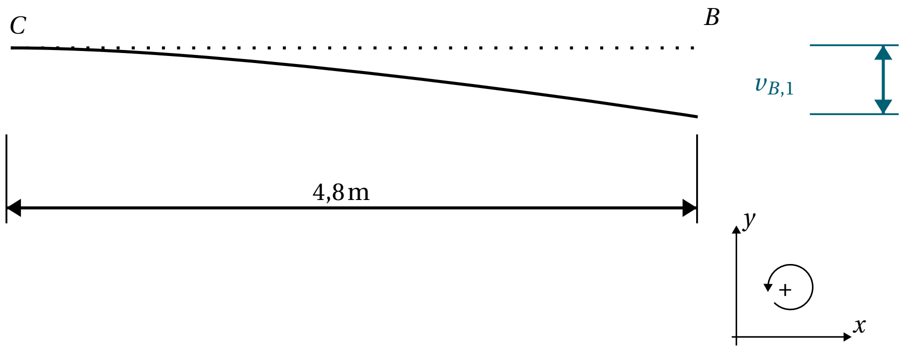


Figure 4.1: Free Body diagram of the cantilevered beam BC.

Derivation

Due to symmetry, it must be true that $R_A = R_B = R_C = -P/2 = 35\,\text{kN}$. As was the case in Problem 16.36 we will once again use the principle of superposition. This time we split the system up into subdivisions consisting of member BC and AB respectively.

Member BC is a cantilevered beam, shown on Figure 4.1. From Appendix D, the maximum displacement of this (which will be at point B) is

$$v_{B,1} = -\frac{\frac{P}{2}L^3}{3EI} = -\frac{PL^3}{6EI}.$$

If we do not consider the deflection caused by the load the member AB will simply rotate rigidly between A and B as depicted on Figure 4.2. I.e. if B "sinks" by a distance of $-PL^3/(6EI)$, then the midpoint D of AB will "sink" half this distance:

$$v_{D,1} = \frac{1}{2}\left(-\frac{PL^3}{6EI}\right) = -\frac{PL^3}{12EI}.$$

For member AB Appendix D gives a deflection of

$$v_{D,\text{load}} = -\frac{PL^3}{48EI}.$$

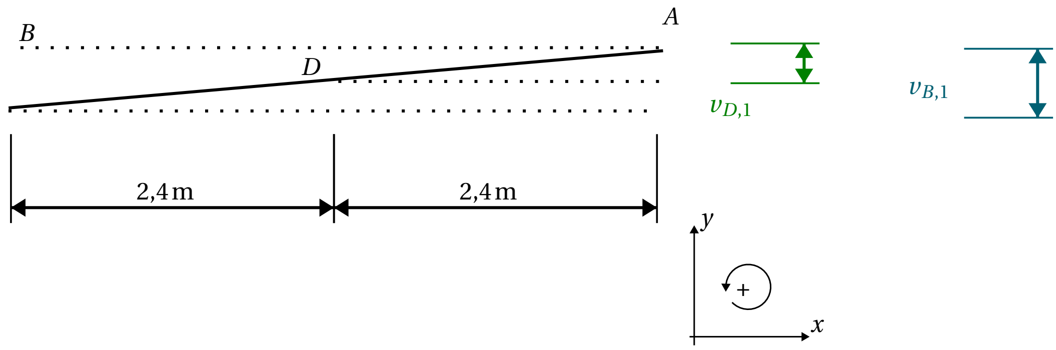


Figure 4.2: Free Body diagram of the rigid rotation of member AB

From the principle of superposition, the total deflection at D will therefore be:

$$v_D = v_{D,1} + v_{D,\text{load}} = -\frac{PL^3}{12EI} - \frac{PL^3}{48EI} = -\frac{5PL^3}{48EI}.$$

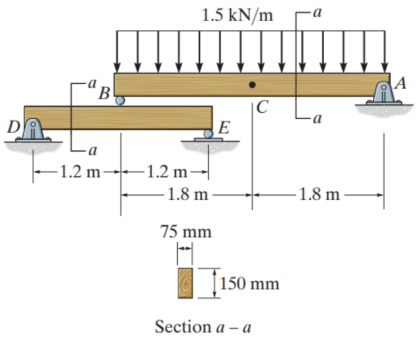
From the figure we see that $P = 70\,\text{kN}$ and $L = 4{,}8\,\text{m}$. Furthermore $I_x = 50 \cdot 10^6\,\text{mm}^4$ is given. In the course literature the elastic modulus of A–36 steel is given as $E = 200\,\text{GPa}$. Hence

$$v_D = -\frac{5 \cdot 70\,\text{kN} \cdot (4{,}8\,\text{m})^3}{48 \cdot 200\,\text{GPa} \cdot 50 \cdot 10^6\,\text{mm}^4} = -8{,}06\,\text{cm}.$$

$$\boxed{v_D = -8{,}06\,\text{cm}}$$ RESULT

Problem R16.7

Using the method of superposition, determine the displacement at C of beam AB. The beams are made of wood having a modulus of elasticity of $E = 10\,\text{GPa}$.



Derivation

Here, we will utilize the same idea of using the principle of superposition as was the case in Problem 16.46.

Due to symmetry we have that

$$R_B = R_A = -\frac{1}{2}W.$$

The equivalent load W that corresponds to our distributed load $w = 1{,}5\,\frac{\text{kN}}{\text{m}}$ is

$$W = w \cdot L_{AB} = -1{,}5\,\frac{\text{kN}}{\text{m}} \cdot 3{,}6\,\text{m} = -5{,}4\,\text{kN}.$$

Hence,

$$R_B = R_A = 2{,}7\,\text{kN}.$$

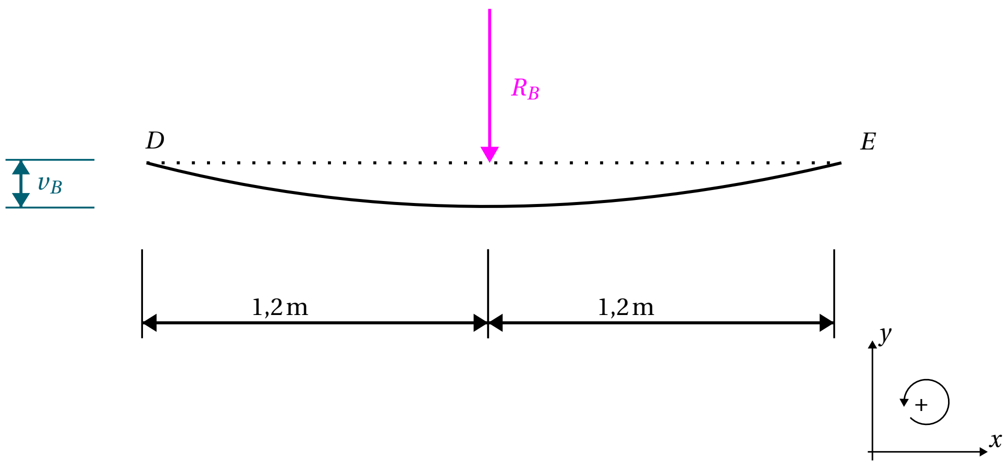


Figure 5.1: Deflection of the simply supported member DE due to the loading at point B

The deflection at the center point B of the simply supported beam DE shown on Figure 5.1 is, per Appendix D:

$$v_B = -\frac{R_B L_{DE}^3}{48EI}.$$

Using the same line of reasoning as was the case in Problem 16.46, the deflection at C will be exactly half of the deflection at B, hence

$$v_{C,1} = \frac{1}{2} v_B = -\frac{R_B L_{DE}^3}{96EI}.$$

The deflection at the midpoint C (the maximum deflection) for the simply supported beam AB is, per Appendix D:

$$v_{C,2} = -\frac{5wL_{AB}^4}{384EI}.$$

The moment of inertia around the neutral axis of each beam is

$$I = \frac{bh^3}{12} = \frac{75\,\text{mm} \cdot (150\,\text{mm})^3}{12} = 2{,}11 \cdot 10^{-5}\,\text{m}^4.$$

Hence, the total displacement becomes:

$$\begin{aligned} v_{C,1} &= -\frac{R_B L_{DE}^3}{96EI} = -\frac{2{,}7\,\text{kN} \cdot (2{,}4\,\text{m})^3}{24 \cdot 10\,\text{GPa} \cdot 2{,}11 \cdot 10^{-5}\,\text{m}^4} = -1{,}84 \cdot 10^{-3}\,\text{m} \\ v_{C,2} &= -\frac{5wL_{AB}^4}{384EI} = -\frac{5 \cdot 1{,}5\,\frac{\text{kN}}{\text{m}} \cdot (3{,}6\,\text{m})^4}{384 \cdot 10\,\text{GPa} \cdot 2{,}11 \cdot 10^{-5}\,\text{m}^4} = -1{,}55 \cdot 10^{-2}\,\text{m} \\ \implies \quad v_C &= v_{C,1} + v_{C,2} = -1{,}84 \cdot 10^{-3}\,\text{m} - 1{,}55 \cdot 10^{-2}\,\text{m} = -17{,}3\,\text{mm}. \end{aligned}$$

$$\boxed{v_C = -17{,}3\,\text{mm}}$$ RESULT